IT24101605 – Fernando C.P.H.A.C
Assigned Model: Random Forest

1. Individual Role and Contributions

- My main responsibility in the GrainWise project was developing and evaluating the Random Forest classification model to distinguish between Cammeo and Osmancik rice varieties. I also contributed to defining the project's quality-control decision-making approach, focusing on how classification could help rice mills identify varieties and reduce incorrect quality-related decisions.
- I developed the Random_Forest.ipynb notebook, including model training, hyperparameter tuning, cross-validation, feature importance analysis, and stability testing. I also saved the model's evaluation results for comparison with the other algorithms developed by my teammates.

2. Technical Learning and Key Decisions

- One of my main learning experiences was understanding how Random Forest combines multiple decision trees to improve classification performance. Unlike distance-based algorithms, Random Forest does not require feature standardisation because its decisions are based on feature thresholds.
- I used GridSearchCV to evaluate 36 hyperparameter combinations, including the number of trees, maximum depth, minimum samples per leaf, and maximum features. The selected model used 400 trees, maximum depth of 6, minimum leaf size of 10, and sqrt feature selection.
- Using repeated stratified cross-validation, my model achieved 92.74% accuracy, 91.48% F1-score, and 97.82% ROC-AUC. This helped me understand the importance of evaluating models using multiple performance metrics.

3. Challenges and Problem-Solving

- A major challenge was understanding how highly correlated features affect Random Forest feature importance. For example, Area and Convex_Area had a correlation of approximately 0.9989, making individual feature importance difficult to interpret.
- To investigate this, I compared impurity-based and permutation-based feature importance. The analysis showed that Major_Axis_Length was the most influential feature, followed by Perimeter in the permutation importance results.
- I also explored model complexity during hyperparameter tuning. The selected model favoured a limited tree depth and larger leaf size, demonstrating how controlling complexity can help reduce overfitting.

4. Critical Evaluation and Reflection

- Comparing Random Forest with Logistic Regression, SVM, and KNN helped me understand that more complex models do not necessarily produce better results. Although Random Forest achieved strong predictive performance, Logistic Regression performed slightly better, and the differences were not statistically significant.
- Testing the Random Forest model with different random seeds also produced consistent results, helping me understand the importance of model stability and reliable validation.

5. Personal Growth and Future Improvements

- This project improved my understanding of ensemble learning, hyperparameter optimization, feature importance, and evidence-based model evaluation. I learned that selecting an appropriate model involves balancing prediction performance, complexity, interpretability, and practical requirements.
- In future projects, I would explore additional feature-selection methods and evaluate the model using rice samples collected under different imaging conditions to improve its real-world reliability.